

## Joint tests and interpretation

**Slide route:** Lecture6v2.tex: single coefficients, joint hypotheses, F tests, linear restrictions and confidence sets. Chapter 7.

### Session 8 | A claim about several coefficients

**Outcomes:** Distinguish separate from joint tests, calculate a classical F statistic, test a linear combination, and interpret restrictions in context.

#### 100-minute route

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|--------|--------------------------------------------------|
| 0–10   | Partial-coefficient retrieval.                   |
| 10–25  | Individual versus joint hypotheses.              |
| 25–45  | F-statistic and restricted/unrestricted example. |
| 45–55  | Robust versus classical tests and pause.         |
| 55–75  | Single restrictions on multiple coefficients.    |
| 75–90  | Confidence sets and model specification.         |
| 90–100 | Exit ticket and functional-form preview.         |

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#### Slide-by-slide script

##### Slide 1 | Title

**Guide** Introduce the class as learning how to test a claim about several coefficients at once.

##### Slide 2 | Roadmap

**Guide** Preview individual tests, joint hypotheses, the F statistic, restrictions, and confidence sets.

##### Slide 3 | Last lesson

**Guide** Retrieve the meaning of a partial coefficient and ask why one coefficient test may not answer a policy question.

##### Slide 4 | Last lesson

**Guide** Use the dummy-variable example to remind students that a group comparison may involve more than one coefficient.

##### Slide 5 | Single coefficients

**Guide** Distinguish  $H_0 : \beta_j = 0$  from a joint claim about several  $\beta$ s.

##### Slide 6 | Joint hypotheses

**Guide** Write  $H_0 : \beta_2 = \beta_3 = 0$  before calculating. Ask what the restricted model removes.

## Slide 7 | Joint hypotheses

**Guide** Explain why separate 5 percent tests do not implement one joint 5 percent test.

## Slide 8 | Joint hypotheses

**Guide** Use the restricted and unrestricted residual sums of squares as the two objects in the F calculation.

## Slide 9 | F statistic

**Guide** Write  $F = ((SSR_R - SSR_U)/q)/(SSR_U/(n - k - 1))$  and label numerator and denominator.

## Slide 10 | F statistic

**Guide** With 120 and 100,  $q = 2$ ,  $n = 100$ ,  $k = 3$ , compute  $F = 9.6$ .

## Slide 11 | F distribution

**Guide** Explain that the reference distribution depends on the restrictions and denominator degrees of freedom.

## Slide 12 | F distribution

**Guide** Ask what a large F means: the restrictions worsen fit relative to residual noise, under the maintained assumptions.

## Slide 13 | Single coefficients

**Guide** Return to a t test and explain that it addresses one restriction.

## Slide 14 | Homoskedastic-only F

**Guide** State that the residual-sum-of-squares formula is not automatically robust.

## Slide 15 | Homoskedastic-only F

**Guide** Use the displayed regression output to identify  $q$ ,  $k$ , and the relevant SSRs.

## Slide 16 | Example with class-size variables

**Guide** Work the deck's joint test and ask students to state the conclusion without overclaiming each coefficient.

## Slide 17 | Example continued

**Guide** Compare the joint conclusion with individual p-values. They need not tell the same story.

## Slide 18 | Why the F statistic?

**Guide** Explain that one statistic aggregates how far the restrictions move the fit.

## Slide 19 | Summary

**Guide** Have students say when a joint test is more natural than a list of individual tests.

## Slide 20 | Single restrictions

**Guide** Introduce  $H_0 : \beta_1 = \beta_2$  as one restriction,  $\beta_1 - \beta_2 = 0$ .

## Slide 21 | Single restrictions

**Guide** Use  $\text{Var}(\hat{\beta}_1 - \hat{\beta}_2)$  and emphasize the covariance term.

## Slide 22 | Rearranging

**Guide** Show that rewriting a restriction changes notation, not the hypothesis.

## Slide 23 | Direct testing

**Guide** Use a software command only after the hand calculation is clear.

## Slide 24 | Confidence sets

**Guide** Explain that a joint confidence region must account for coefficient covariance.

## Slide 25 | Confidence sets

**Guide** Contrast a confidence ellipse with the Cartesian product of two individual intervals.

## Slide 26 | Confidence sets

**Guide** Ask which parameter combinations are ruled out by the region.

## Slide 27 | Confidence sets

**Guide** Use the interaction preview: a conditional effect is a linear combination that may need its own interval.

## Slide 28 | Specification

**Guide** Ask which variables belong in a model based on the question and design, not only on significance.

## Slide 29 | Specification

**Guide** Warn that dropping controls because they are individually insignificant can reintroduce OVB.

## Slide 30 | Summary

**Guide** Review joint null, restriction count, covariance, and reference distribution.

## Slide 31 | Material

**Guide** Assign one robust Wald exercise and one equality restriction before the functional-form class.

## Worked example | Two exclusions

**Use with slides** Lecture6v2 slides 7–16: Tests of Joint Hypotheses through the example on *strs* and *frpmfracs*. Use the SSR calculation when slide 11 introduces the F-statistic, then compare it with the deck’s example on slide 16.

Test  $H_0 : \beta_2 = \beta_3 = 0$ . A restricted regression has residual sum of squares 120; the unrestricted regression has 100. Let  $n = 100$ , unrestricted slope count  $k = 3$ , and restriction count  $q = 2$ .

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} = \frac{(120 - 100)/2}{100/96} = 9.6.$$

Under homoskedastic normal errors, compare with  $F_{2,96}$ ; its 5% critical value is about 3.09, so reject.

**What to say** The joint claim that both coefficients are zero is inconsistent with the data under these assumptions. This does not establish that each individual coefficient is nonzero.

**Robust caveat** The residual-sum-of-squares formula is the homoskedastic-only test. Under heteroskedasticity, use the estimated robust covariance matrix in a Wald test; do not relabel this formula robust.

## Worked example | Equality is one restriction

**Use with slides** Lecture6v2 slides 22–24: Testing Single Restrictions, Rearranging, and Direct Testing. Use this equality example before the Stata command so students see why the covariance term matters.

Suppose  $\hat{\beta}_1 = .5$ ,  $\hat{\beta}_2 = .2$ , their estimated variances are .04 and .01, and covariance is .01. To test  $H_0 : \beta_1 = \beta_2$ :

$$\widehat{\beta_1 - \beta_2} = .3, \quad SE(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{.04 + .01 - 2(.01)} = \sqrt{.03}.$$

$$t = .3/\sqrt{.03} \simeq 1.732, \quad 95\% \text{ large-sample CI: } .3 \pm 1.96\sqrt{.03} = [-.0395, .6395].$$

Fail to reject equality at 5%. Each coefficient's separate significance relative to zero cannot answer whether the two coefficients differ.

**Pair prompt** Why can we not subtract the two standard errors? Because the variance of a difference includes both variances and their covariance.

**Multiple-testing side example** If two tests were independent and each used 5%, the probability of at least one false rejection under the joint null would be  $1 - .95^2 = .0975$ . Regression coefficient tests need not be independent, but this shows why separate tests do not implement a joint 5% test.

**Software illustration** With an appropriately fitted Stata model:

```
test x2 x3
test x1 = x2
```

Use the model's robust covariance estimate when robust inference is required. Confirm variable names before demonstrating.

**Confidence sets** A joint confidence ellipse incorporates coefficient covariance. It is not obtained by declaring the Cartesian product of individual 95% intervals a joint 95% region.

**Exit answers** For one restriction, a consistently constructed Wald F equals the corresponding  $t^2$ . Equality of two coefficients is one restriction. An insignificant individual test does not preclude joint significance.

**If short / preparation** Leave repeated output screenshots for reading. Preserve both worked examples. Read Chapter 8 and ask students to sketch a relationship whose slope changes with  $X$ .

**Source:** Lecture6v2.tex; synthetic numerical examples.